

Sample End Semester Question Paper - 2025

Subject: Strength of Materials | Code: ME301 | Semester: 3

Branch: Mechanical | Regulation: R20 (sample) | Time: 3 Hours | Max I

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. What is a column?
2. Define slenderness ratio.
3. Name one theory of failure.
4. What is a point of contraflexure?
5. Define Young's modulus.
6. Define stress and strain.
7. State Hooke's law.
8. What is a shear force diagram?
9. Define section modulus.
10. State the torsion equation.

PART B - Long questions (6 x 13 = 78 Marks)

1. Explain the stress-strain diagram. Derive the relation between E, G, K and Poisson's ratio.
2. Draw SFD and BMD for a given simply supported beam. Find maximum bending stress.
3. Derive the bending equation. Find section modulus of a given section.
4. Derive the torsion equation. Solve a circular shaft problem.
5. Explain Euler's column theory. Find the critical load for a given column.
6. Explain theories of failure: maximum principal stress and maximum shear stress with examples.